

St George Girls High School

Year 12

Assessment Task 3

2006



Mathematics

Extension 1

General Instructions

- Time allowed – 75 minutes
- Write using blue or black pen
- Board-approved calculators may be used.
- A table of standard integrals is provided.
- All necessary working should be shown in every question.
- Write on one side of the page only.
- Start each question on a new page.

Total marks – 70

- Attempt Questions 1 – 5
- All questions are of equal value

Question	Mark
Question 1	/14
Question 2	/14
Question 3	/14
Question 4	/14
Question 5	/14
Total	/70

Question 1 – (14 marks) – Start a new page

Marks

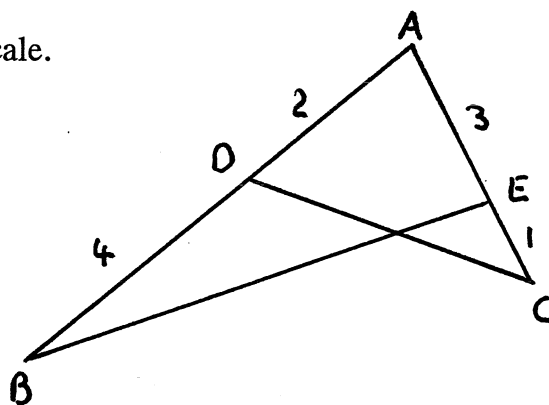
- a) Find the value of 'a' such that $P(x) = x^3 - 2x^2 - ax + 6$ is divisible by $(x + 2)$

2

- b) The diagram opposite is not drawn to scale.

4

Given $AD = 2; BD = 4$
 $AE = 3; CE = 1$



- (i) Prove that $\triangle ABE \parallel \triangle ACD$

- (ii) If $BE = 5$, find the length of DC

- c) Find $\int_0^{\frac{\pi}{2}} \cos^2 \frac{x}{2} dx$

3

- d) If α, β, γ are the roots of the cubic equation $5x^3 - 2x^2 - 3 = 0$, find the value of:

5

(i) $\alpha + \beta + \gamma$

(ii) $\alpha\beta + \alpha\gamma + \beta\gamma$

(iii) $\alpha\beta\gamma$

(iv) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$

Question 2 – (14 marks) – Start a new page

Marks

a) (i) Show that $\sqrt{12} \sin x + 2 \cos x \equiv 4 \cos\left(x - \frac{\pi}{3}\right)$

5

(ii) Hence, solve the equation $\sqrt{12} \sin x + 2 \cos x = -3$ for $0 \leq x \leq 2\pi$

[Give all answers correct to two decimal places]

b) (i) Factorise completely the polynomial $P(x) = x^3 - x^2 - 8x + 12$ given that the equation $P(x) = 0$ has a double root at $x = 2$

3

(ii) When $Q(x) = ax^3 + bx + c$ is divided by $(x - 1)$, the remainder is -4 .

When $Q(x) = ax^3 + bx + c$ is divided by $(x^2 - 4)$, the remainder is $(-4x + 3)$

4

Find a , b and c .

c) A particle moving in a straight line has its acceleration as a function of time given by $\ddot{x} = 3t - 1$

If the particle is initially at rest two units to the right of the origin, find its displacement at $t = 3$

2

Question 3 – (14 marks) – Start a new page

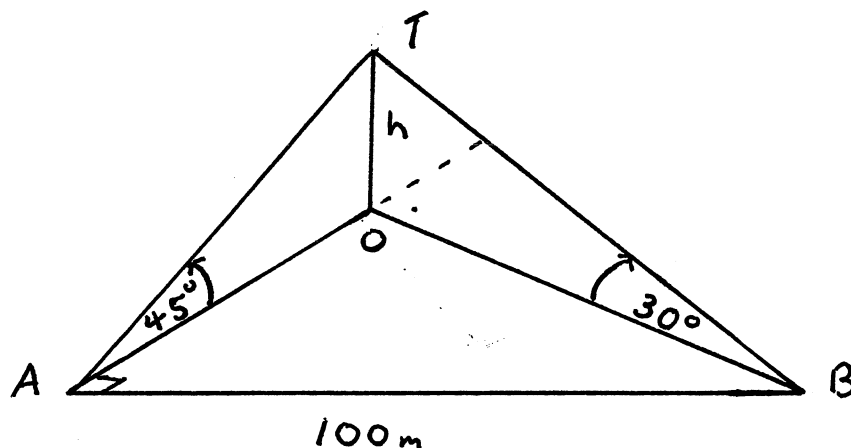
Marks

a) Prove $\frac{\sin 2x}{1 + \cos 2x} = \tan x$

2

b)

6



A surveyor stands at a point A , which is due south of a tower OT of height h metres. The angle of elevation of the top of the tower from A is 45° . The surveyor then walks 100m due east to point B , from where she measures the angle of elevation of the top of the tower to be 30° .

(i) Express the length OB in terms of h .

(ii) Show that $h = 50\sqrt{2}$

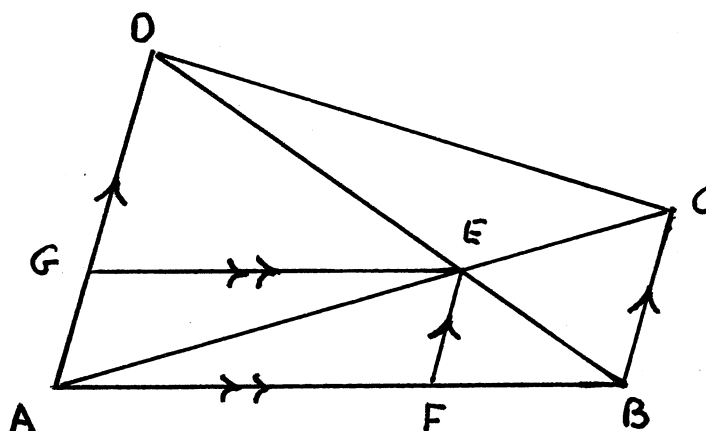
(iii) Calculate the bearing of B from the base of the tower.

Question 3 (cont'd)

Marks

c)

6



$ABCD$ is a trapezium in which

(i) $BC \parallel AD$

(ii) $BC : AD = 1 : 2$

(iii) $EF \parallel BC$

(iv) $EG \parallel AF$

(α) Show that $FE : BC = AF : AB$

(β) Show that $DG : AD = GE : AB$

(γ) Show that $FE : BC = DG : AD$

(ϵ) Show that $DG = AD - FE$

Hence, (or otherwise) prove that the ratio $FE : BC = 2 : 3$

Question 4 – (14 marks) – Start a new page

Marks

- a) The polynomials $4x^3 - x^2 + 3$ and $ax(x+1)(x+2) + bx(x+1) + cx + d$ are equal for 4 values of x ; determine the values of a, b, c and d .

4

- b) The velocity $\dot{x} \text{ ms}^{-1}$ of a particle moving in a straight line is given by

$$\dot{x} = \frac{4}{\pi} \sin \frac{\pi t}{2}$$

7

Initially the particle is at rest at the origin.

- (i) Find an expression for displacement as a function of time.
- (ii) Find an expression for acceleration as a function of time.
- (iii) Sketch the graph of $\dot{x}$ as a function of time over $0 \leq t \leq 8$.
- (iv) What is the displacement of the particle each time it is at rest during the period $0 \leq t \leq 8$.

- c) Show that $\sin(A+B) + \sin(A-B) = 2 \sin A \cos B$

3

Hence, find $\int 12 \sin 4x \cos 2x \, dx$

Question 5 – (14 marks) – Start a new page

Marks

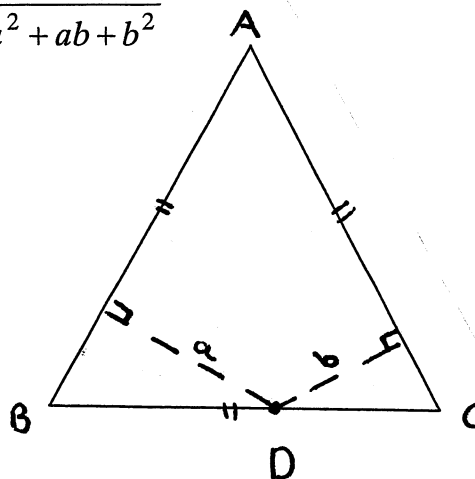
a) (i) Write down an expression for $\sin(x - y)$.

6

(ii) Given $\sin \alpha = c$ and $\sin(60 - \alpha) = d$ prove that $c^2 + cd + d^2 = \frac{3}{4}$

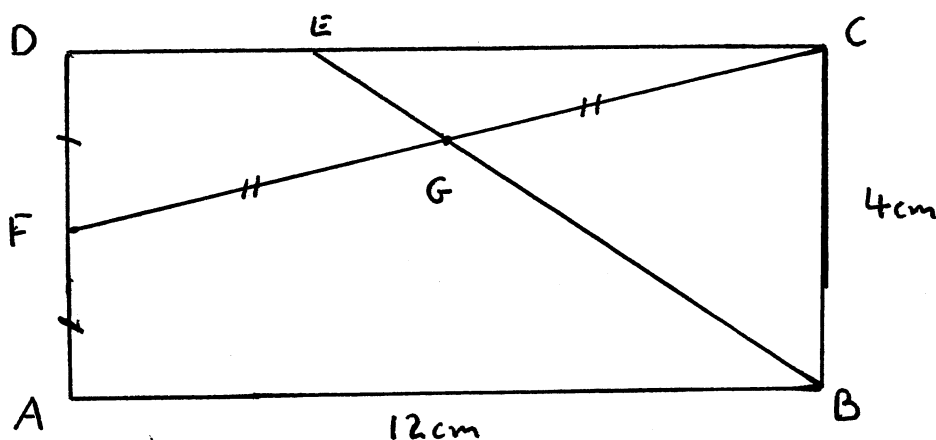
(iii) $\triangle ABC$ is equilateral and D is any point on the side BC . The lengths of the perpendiculars from D to AB and AC are a and b respectively.

Prove that $AD = \frac{2}{\sqrt{3}} \sqrt{a^2 + ab + b^2}$



b)

[not to scale]



In the figure, $ABCD$ is a rectangle. $AB = 12\text{cm}$ and $BC = 4\text{cm}$.
 F is the midpoint of AD and G is the midpoint of CF .

Find the length of DE . [Hint: Construct perpendicular from G to X on DC].

c) If $a \cos x = 1 + \sin x$ prove that $\frac{a-1}{a+1} = t$, where $t = \tan \frac{x}{2}$

3

TABLE OF STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

Note $\ln x = \log_e x, \quad x > 0$

QUESTION 1:

(a) $P(x) = x^3 - 2x^2 - ax + 6$
 $P(-2) = 0 \Rightarrow -8 - 8 + 2a + 6 = 0$
 $2a = 10$
 $\therefore a = 5$

(b) (i) In $\triangle ABE$, $\triangle ACD$

(i) $\hat{A}$ is common

(ii) $\frac{AB}{AC} = \frac{3}{2} = \frac{AE}{AD}$

$\therefore \triangle ABE \parallel \triangle ACD$ (one angle equal and the sides about that angle are in same ratio)

(ii) $\frac{5}{x} = \frac{3}{2} \Rightarrow 3x = 10$
 $x = \frac{10}{3}$

(c) $\int_0^{\frac{\pi}{2}} \cos^2\left(\frac{x}{2}\right) dx = \int_0^{\frac{\pi}{2}} \frac{\cos x + 1}{2} dx$
 $= \frac{1}{2} \left[\sin x + x \right]_0^{\frac{\pi}{2}}$
 $= \frac{1}{2} \left[1 + \frac{\pi}{2} - 0 \right]$
 $= \frac{\pi + 2}{4}$

(d) (i) $\angle + \beta + \gamma = \frac{2}{5}$ (ii) $\angle\beta + \angle\gamma + \beta\gamma = 0$

(iii) $\angle\beta\gamma = \frac{3}{5}$ (iv) $\frac{1}{\angle} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\angle\beta + \angle\gamma + \beta\gamma}{\angle\beta\gamma}$

$= 0$

QUESTION 2:

$$\begin{aligned} \text{(a) (i)} \quad 4 \cos\left(x - \frac{\pi}{3}\right) &= 4 \cos x \cos \frac{\pi}{3} + 4 \sin x \sin \frac{\pi}{3} \\ &= 2 \cos x + 4 \sin x \cdot \frac{\sqrt{3}}{2} \\ &= 2\sqrt{3} \sin x + 2 \cos x \\ &= \sqrt{12} \sin x + 2 \cos x. \end{aligned}$$

$$\text{(ii)} \quad 4 \cos\left(x - \frac{\pi}{3}\right) = -3$$

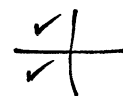
$$\Rightarrow \cos\left(x - \frac{\pi}{3}\right) = -\frac{3}{4}$$

$$\therefore x - \frac{\pi}{3} = \pi - \alpha, \pi + \alpha$$

$$x = \frac{4\pi}{3} - \alpha, \frac{4\pi}{3} + \alpha$$

$$\begin{aligned} 0 &\leq x \leq 2\pi \\ -\frac{\pi}{3} &\leq x - \frac{\pi}{3} \leq \frac{5\pi}{3} \end{aligned}$$

$$\left(x - \frac{\pi}{3}\right)_{\text{acute}} = 0.7227..$$



$$= 3.47, 4.91$$

$$\text{(b) (i)} \quad P(x) = x^3 - x^2 - 8x + 12$$

Let roots of $P(x) = 0$ be $2, 2, \alpha$

$$\begin{aligned} \sum \text{roots} &= 1 \Rightarrow 4 + \alpha = 1 \\ \alpha &= -3 \end{aligned}$$

$$\therefore P(x) = (x-2)(x-2)(x+3)$$

$$\text{(ii)} \quad Q(x) = ax^3 + bx + c$$

$$Q(1) = -4 \Rightarrow a + b + c = -4 \quad \text{--- (1)}$$

$$Q(x) = (x^2 - 4) \cdot P(x) + (-4x + 3)$$

$$\therefore Q(2) = -5 \Rightarrow 8a + 2b + c = -5 \quad \text{--- (2)}$$

$$Q(-2) = 11 \Rightarrow -8a - 2b + c = 11 \quad \text{--- (3)}$$

$$\begin{aligned} \text{(2)} + \text{(3)}: \quad 2c &= 6 \\ c &= 3 \text{ sub in (1), (2)} \end{aligned}$$

$$\Rightarrow a + b = -7 \quad \text{--- (4)}$$

$$8a + 2b = -8 \quad \text{--- (5)}$$

$$\text{(5)} - 2 \times \text{(4)}: 6a = 6 \quad \therefore a = 1$$

(c)

$$\ddot{x} = 3t - 1$$

$$\Rightarrow \frac{dv}{dt} = 3t - 1$$

$$\therefore v = \frac{3t^2}{2} - t + C$$

$$\left. \begin{matrix} t=0 \\ v=0 \end{matrix} \right\} \Rightarrow 0 = 0 + C \quad \therefore C = 0$$

$$\therefore \frac{dx}{dt} = \frac{3t^2}{2} - t$$

$$\therefore x = \frac{t^3}{2} - \frac{t^2}{2} + C$$

$$\left. \begin{matrix} t=0 \\ x=2 \end{matrix} \right\} \Rightarrow 2 = C$$

$$\therefore x = \frac{t^3}{2} - \frac{t^2}{2} + 2$$

$$\text{at } t=3 \quad x = \frac{27}{2} - \frac{9}{2} + 2$$

$$= 11$$

QUESTION 3:

$$\begin{aligned} \text{(a)} \quad \text{LHS} &= \frac{\sin 2x}{1 + \cos 2x} \\ &= \frac{2 \sin x \cos x}{1 + (2 \cos^2 x - 1)} \\ &= \frac{2 \sin x \cos x}{2 \cos^2 x} \\ &= \tan x \\ &= \text{RHS} \end{aligned}$$

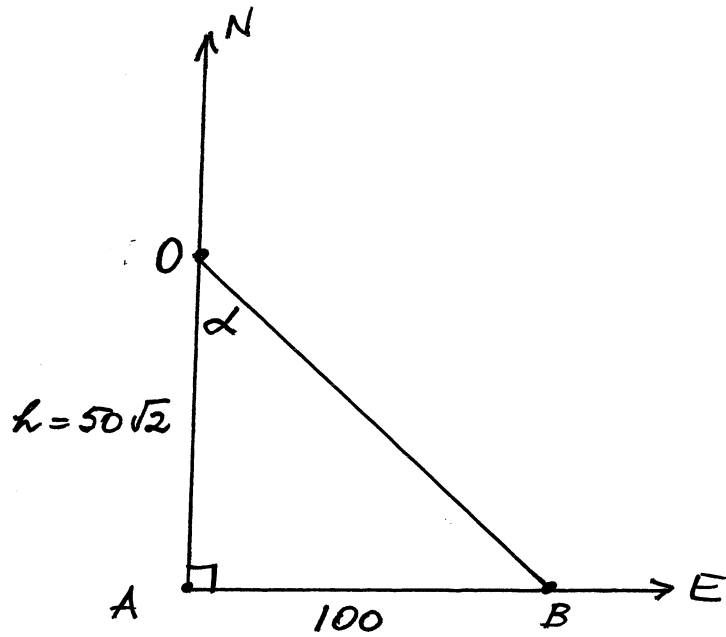
$$\therefore \frac{\sin 2x}{1 + \cos 2x} = \tan x$$

$$\begin{aligned} \text{(b)} \quad \text{(i)} \quad \text{In } \triangle OBT: \tan 60^\circ &= \frac{OB}{L} \\ \therefore OB &= L \tan 60^\circ \\ &= L\sqrt{3} \end{aligned}$$

$$\text{(ii)} \quad \text{In } \triangle OAT: \quad OA = L \quad (\text{isosceles } \triangle OAT, \quad \angle AOT = \angle OTA = 45^\circ)$$

$$\begin{aligned} \text{In } \triangle OAB: \quad OB^2 &= OA^2 + 100^2 \\ \therefore (L\sqrt{3})^2 &= L^2 + 100^2 \\ 3L^2 &= L^2 + 10000 \\ 2L^2 &= 10000 \\ L^2 &= 5000 \\ L &= \sqrt{5000} \\ &= 50\sqrt{2} \end{aligned}$$

(iii) Birds eye view



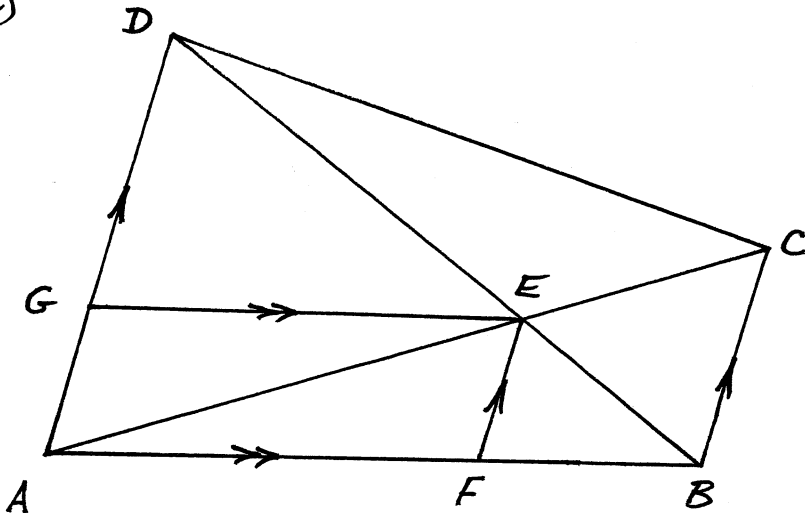
$$\text{In } \triangle OAB: \tan \hat{AOB} = \frac{100}{50\sqrt{2}}$$

$$= \sqrt{2}$$

$$\therefore \hat{AOB} = 54^{\circ} 44'$$

$\therefore$ Bearing of B from base of tower is $125^{\circ} 16'$

(c)



(c)

(α) In $\triangle AEF, \triangle ACB$

• $\hat{A}$ is common

• $\hat{AFE} = \hat{ACB}$ (corresponding angles equal, $FE \parallel BC$)

• $\hat{AEF} = \hat{ABC}$ (... ..)

$\therefore \triangle AEF \parallel \triangle ACB$ (equiangular)

$\therefore FE : BC = AF : AB$ (corresponding sides of similar triangles in same ratio)

(β) Similarly $\triangle DGE \parallel \triangle DAB$

$\therefore DG : AD = GE : AB$ (corresponding sides of similar triangles in same ratio)

(γ) now $\frac{FE}{BC} = \frac{AF}{AB}$

$= \frac{GE}{AB}$ since $AF = GE$ (opposite sides of parallelogram are equal)

$= \frac{DG}{AD}$ from (β)

QUESTION 4:

$$(a) \quad 4x^3 - x^2 + 3 \equiv ax(x+1)(x+2) + bx(x+1) + cx + d$$

since the two polynomials are equal for 4 values of x .

$$x=0 \Rightarrow 3 = d$$

$$x=-1 \Rightarrow -2 = -c + d \\ = -c + 3$$

$$\therefore c = 5$$

$$\text{coeff of } x^3 \Rightarrow 4 = a$$

$$x=-2 \Rightarrow -33 = -2b(-1) - 2c + d$$

$$-33 = 2b - 10 + 3$$

$$\therefore 2b = -26$$

$$\therefore b = -13$$

$$\therefore a = 4, b = -13, c = 5, d = 3$$

$$(b) \quad \ddot{x} = \frac{4}{\pi} \sin \frac{\pi t}{2}$$

$$(i) \quad x = \frac{-\frac{4}{\pi} \cdot \cos \frac{\pi t}{2}}{\left(\frac{\pi}{2}\right)} + C$$

$$x = -\frac{8}{\pi^2} \cos \frac{\pi t}{2} + C$$

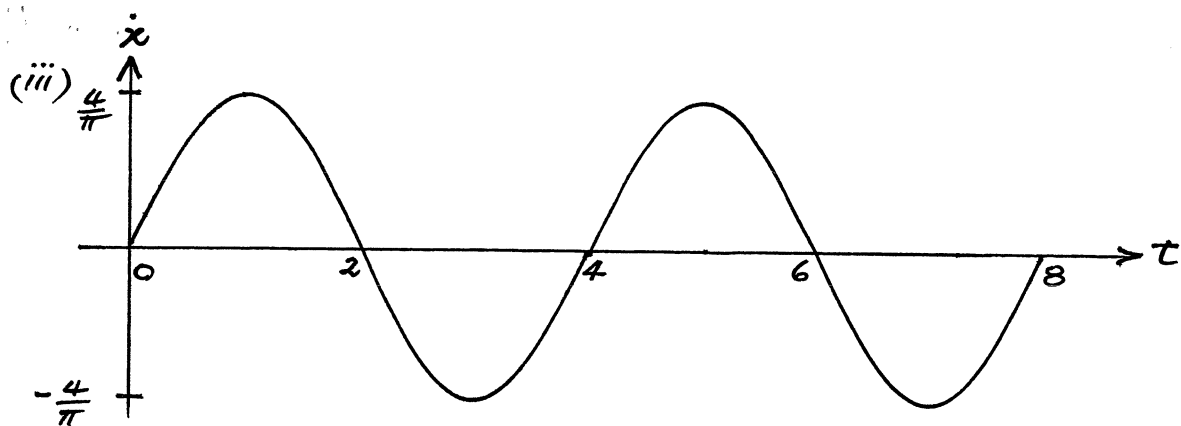
$$\text{at } t=0, x=0 \therefore 0 = -\frac{8}{\pi^2} + C$$

$$\therefore C = \frac{8}{\pi^2}$$

$$\therefore x = \frac{8}{\pi^2} - \frac{8}{\pi^2} \cos \frac{\pi t}{2}$$

$$(ii) \quad \ddot{x} = \frac{4}{\pi} \cdot \frac{\pi}{2} \cos \frac{\pi t}{2}$$

$$= 2 \cos \frac{\pi t}{2}$$



(iv) Particle at rest at $x = 0$

i.e. $t = 0, 2, 4, 6, 8$ sub in ①

$$\left\{ \begin{array}{l} t=0 \\ x=0 \end{array} \right\} \quad \left\{ \begin{array}{l} t=2 \\ x=\frac{16}{\pi^2} \end{array} \right\} \quad \left\{ \begin{array}{l} t=4 \\ x=0 \end{array} \right\} \quad \left\{ \begin{array}{l} t=6 \\ x=\frac{16}{\pi^2} \end{array} \right\} \quad \left\{ \begin{array}{l} t=8 \\ x=0 \end{array} \right\}$$

(c) $\sin(A+B) + \sin(A-B)$

$$= \sin A \cos B + \cos A \sin B + \sin A \cos B - \cos A \sin B$$

$$= 2 \sin A \cos B \quad \text{————— ①}$$

$$\int 12 \sin 4x \cos 2x \, dx$$

$$= 6 \int 2 \sin 4x \cos 2x \, dx$$

$$= 6 \int (\sin 6x + \sin 2x) \, dx \text{ from ①}$$

$$= 6 \left[-\frac{1}{6} \cos 6x - \frac{1}{2} \cos 2x \right] + C$$

$$= -\cos 6x - 3 \cos 2x + C$$

QUESTION 5:

(a) (i) $\sin(x-y) = \sin x \cos y - \cos x \sin y$

(ii) $\sin \angle = c$ } ——— ①
 $\sin(60^\circ - \angle) = d$ } ——— ②

from ②: $d = \sin 60^\circ \cos \angle - \cos 60^\circ \sin \angle$
 $= \frac{\sqrt{3}}{2} \cos \angle - \frac{1}{2} \sin \angle$

$$d = \frac{\sqrt{3}}{2} \cos \angle - \frac{c}{2}$$

$$\Rightarrow 2d = \sqrt{3} \cos \angle - c$$

$$\Rightarrow \cos \angle = \frac{2d+c}{\sqrt{3}}$$

Then $\sin^2 \angle + \cos^2 \angle = 1$

$$\Rightarrow c^2 + \left(\frac{2d+c}{\sqrt{3}}\right)^2 = 1$$

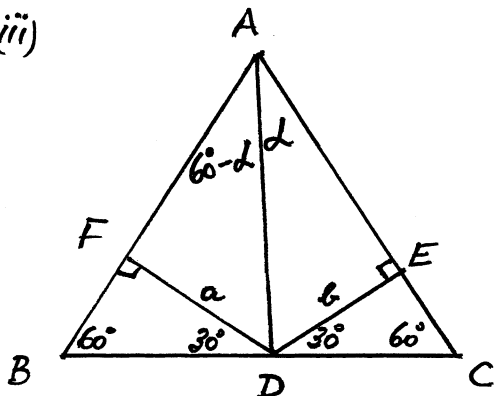
$$c^2 + \frac{4d^2 + 4cd + c^2}{3} = 1$$

$$3c^2 + 4d^2 + 4cd + c^2 = 3$$

$$4c^2 + 4cd + 4d^2 = 3$$

$$\therefore c^2 + cd + d^2 = \frac{3}{4}$$

(iii)



Let $\hat{DAC} = \angle$

$$\hat{DAB} = 60^\circ - \angle$$

In $\triangle ADE$: $\sin \angle = \frac{b}{AD}$

$$\triangle ADF: \sin(60^\circ - \angle) = \frac{a}{AD}$$

(ii) let $c = \frac{b}{AD}$ and $d = \frac{a}{AD}$

Then $c^2 + cd + d^2 = \frac{3}{4}$

$$\Rightarrow \frac{b^2}{AD^2} + \frac{ab}{AD^2} + \frac{a^2}{AD^2} = \frac{3}{4}$$

$$\Rightarrow b^2 + ab + a^2 = \frac{3}{4} \cdot AD^2$$

$$\therefore AD = \frac{2}{\sqrt{3}} (b^2 + ab + a^2)^{\frac{1}{2}}$$

(b) Construct $GX \perp DC$ at X .

$CX = XD = 6 \text{ cm}$ (parallel lines AD , GX and BC cut intercepts in same ratio and $FG = GC$)

$$GX = \frac{1}{2} DF = 1$$

(The line joining the mid-points of two sides of $\triangle CDF$ is parallel to the third side and equal to half its length)

OR prove the same by similar triangles.

Let $DE = x$

Hence $EX = 6 - x$

In $\triangle EXG$, $\triangle ECB$

(i) $\hat{E}$ is common

(ii) $\hat{EXG} = \hat{ECB} = 90^\circ$ (by construction and angles of a rectangle)

(iii) $\hat{EGX} = \hat{ECB}$ (angle sum of triangle is 180°)

$\therefore \triangle EXG \parallel \triangle ECB$ (equiangular)

$$\therefore \frac{EX}{EC} = \frac{XG}{CB} \quad (\text{corresponding sides of similar triangles are in the same ratio})$$

$$\frac{6-x}{12-x} = \frac{1}{4}$$

$$\Rightarrow x = 4 \quad \therefore DE = 4 \text{ cm}$$

$$(c) \quad a \cos x = 1 + \sin x$$

$$\Rightarrow a = \sec x + \tan x$$

$$\frac{a-1}{a+1} = \frac{\sec x + \tan x - 1}{\sec x + \tan x + 1}$$

$$= \frac{1 + \sin x - \cos x}{1 + \sin x + \cos x}$$

$$= \frac{1 + \frac{2t}{1+t^2} - \frac{1-t^2}{1+t^2}}{1 + \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}} \quad t \equiv \tan \frac{x}{2}$$

$$= \frac{1+t^2 + 2t - 1 + t^2}{1+t^2 + 2t + 1 - t^2}$$

$$= \frac{2t^2 + 2t}{2t + 2}$$

$$= \frac{2t(t+1)}{2(t+1)}$$

$$= t \quad \text{where } t \equiv \tan \frac{x}{2}$$
